

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

PH1003-1 – PHYSICS OF FLUIDS AND SEMICONDUCTORS
(BT)

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

List of constants: Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$, Planck's Constant, $h = 6.63 \times 10^{-34} \text{ Js}$, Electron mass, $m = 9.1 \times 10^{-31} \text{ Kg}$, Electron Charge, $e = 1.6 \times 10^{-19} \text{ C}$, Boltzmann Constant, $k = 1.38 \times 10^{-23} \text{ J/K}$, Avogadro Number, $N_A = 6.023 \times 10^{26} / \text{kg Mole}$, Density of water = 1000 kgm^{-3} .

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The following don't have zero shear stress
 A) Liquids
 B) Gases
 C) Compressible fluids
 D) Solids
- A fluid is flowing steadily through a pipe. At which point in the pipe is the fluid (non-ideal) velocity the highest?
 A) At the center of the pipe
 B) Near the walls of the pipe
 C) It's the same throughout the pipe
 D) At the inlet of the pipe
- There is a bucket of water with a hole at the bottom: the velocity of the water flowing out as time elapses
 A) Increases
 B) Decreases
 C) Remains same
 D) Initially increases later decreases
- Viscosity is
 A) Pressure dependent
 B) Velocity dependent
 C) Density dependent
 D) Velocity independent
- When an object is partially or fully immersed in a liquid its apparent weight is
 A) Equal to the weight of the fluid displaced
 B) Same as its weight in the air
 C) Buoyant force
 D) The difference between the weight of the object in the air and the buoyant force
- What phenomenon is responsible for the lift force generated on an airfoil?
 A) Viscosity
 B) Pressure difference
 C) Turbulence
 D) Drag force
- When a solid ball fall through a highly viscous liquid in a long jar, its velocity
 A) Will keep increasing due to gravity
 B) Will initially increase then keep on decreasing till attains a constant velocity
 C) Will decrease as soon as it starts moving
 D) Will initially increase and keep decreasing without attaining a final constant velocity
- What effect does increasing the viscosity of a fluid have on the Reynolds number of the flow?
 A) Increases
 B) Remains constant
 C) Decreases
 D) Depends on fluid viscosity
- Classical free electron theory is based on the assumption that
 A) Electrons are freely moving only at the boundary region of the solid
 B) Electrons are freely moving through the entire solid
 C) Electrons obey Pauli exclusion principle
 D) Electrons can only have discrete energy levels
- The mobility of electrons in copper is $3.5 \times 10^{-3} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$. For an applied electric field strength of 2 V/m , the drift velocity of the free electrons is
 A) $0.7 \times 10^{-3} \text{ m/s}$
 B) $7 \times 10^{-3} \text{ m/s}$
 C) $1.75 \times 10^{-3} \text{ m/s}$
 D) $7 \times 10^3 \text{ m/s}$
- Fermi level for a metal is
 A) Highest energy level occupied by electrons at 0 K
 B) Average value of all available energy levels
 C) Highest energy level occupied by electrons at 0° C
 D) Addition of energy of all available electron energy levels

12. According to Quantum free electron theory: The flow of electrons is not affected by
 A) Thermal vibrations B) Impurity atoms
 C) Crystal defects D) Collisions with ion cores
13. Which of the following statement correctly describes a metal based on band theory?
 A) A material possessing a moderate band gap B) A material possessing a large band gap
 C) A material with an infinite band gap D) A material with zero band gap
14. When an n-type semiconductor is attached to a p-type semiconductor the fermi level in the n-type will be
 A) At the same level that of the fermi level at the p-type B) Lower than the fermi level of the p-type semiconductor
 C) Higher than the fermi level of the p-type semiconductor D) Higher if the doping on the n-type semiconductor is higher.
15. Hall effect is
 A) The generation of an electric potential difference perpendicular to both an electric current and a magnetic field in a conductor. B) The increase in electrical conductivity of a material when subjected to a magnetic field.
 C) The emission of photons when electrons transition between energy levels in a semiconductor. D) The change in temperature of a material when an electric current is passed through it.
16. An LED emits light with a bandwidth of 33THz (Terahertz). If the frequency of the emitted light is 5.0×10^{14} Hz. Given that the speed of light in a vacuum is 3.00×10^8 meters per second, the range of wavelengths of this light in nanometers (nm) is:
 A) $400\text{nm} \leq \lambda \leq 600\text{nm}$ B) $600\text{nm} \leq \lambda \leq 800\text{nm}$
 C) $500\text{nm} \leq \lambda \leq 700\text{nm}$ D) $300\text{nm} \leq \lambda \leq 500\text{nm}$
17. A charged particle with a charge of 2×10^{-6} C experiences a force of 4×10^{-3} N in an electric field. What is the magnitude of the electric field strength at the location of the particle?
 A) $2 \times 10^3 \text{ N/C}$ B) $2 \times 10^{-3} \text{ N/C}$
 C) $8 \times 10^{-4} \text{ N/C}$ D) $8 \times 10^3 \text{ N/C}$
18. In mass spectrometry, what is the purpose of the mass analyzer?
 A) To ionize molecules B) To detect ions
 C) To separate ions based on their mass-to-charge ratio D) To accelerate ions
19. Which method is utilized in X-ray crystallography to determine the atomic structure of crystalline materials?
 A) X-ray fluorescence B) X-ray absorption spectroscopy
 C) X-ray diffraction D) X-ray photoelectron spectroscopy
20. In scanning electron microscopy (SEM), what is the primary role of the electron detector?
 A) To produce electron beams B) To magnify the image
 C) To collect secondary electrons D) To focus the electron beam

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Derive the expression for pressure of a fluid under static equilibrium at different depths, showing the force diagrams.	8	L*3	1	1,2
	b) Why does the raindrop not hurt when it falls on us from such a great height? Explain in detail.	4	L3	2	1,2
	c) A ball of mass 2 kg having a diameter of 50 cm falls in the swimming pool. Calculate its buoyant force and volume of water displaced.	4	L3	1	1,2
2.	a) Explain the principle of manometry. With a neat diagram explain how manometry is carried out.	6	L2	1	1,2
	b) Using Bernoulli's equation explain two applications of Bernoulli's equation in detail.	4	L3	2	1,2
	c) A crown is suspected to have a metal impurity instead of pure gold, using the principle of fluid statics how would you confirm if it has an impurity or not? Explain the logic and the steps involved. (Note: Suggest a practical method which is doable)	6	L3	1	1,2

- | | | | | | |
|-------|--|---|----|---|-----|
| 3. a) | Derive an expression for the viscosity of a liquid flowing on a surface with a neat diagram. Discuss the result and its significance. | 8 | L3 | 1 | 1,2 |
| b) | Describe the difference between laminar and turbulent flow. | 4 | L2 | 2 | 1,2 |
| c) | A metal plate of area $2.5 \times 10^{-4} \text{ m}^2$ is placed on a $0.25 \times 10^{-3} \text{ m}$ thick layer of castor oil. If a force of 2.5 N is needed to move the plate with a velocity $3 \times 10^{-2} \text{ m s}^{-1}$, calculate the coefficient of viscosity of castor oil. | 4 | L3 | 2 | 1,2 |

Unit – II

- | | | | | | |
|-------|--|---|----|---|-----|
| 4. a) | Explain Drude's model and define i) Mean free path ii) Collision time and explain how it explains the origin of resistance. | 6 | L2 | 3 | 1,2 |
| b) | Explain the main features of classical free electron theory. Explain the any two main differences of Classical and Quantum free electron theory. | 6 | L2 | 3 | 1,2 |
| c) | A copper wire is 1 m long and has a cross-section of 0.1 mm^2 , the resistance of the wire at room temperature is found to be 0.72 ohm. What is the resistivity of the material? | 4 | L3 | 3 | 1,2 |
| 5. a) | State Matthiessen's rule. How does it explain the origin of resistivity differently than classical free electron theory? | 3 | L3 | 3 | 1,2 |
| b) | A current is flowing in a rectangle semiconductor slab. By applying a magnetic field on it, how Hall voltage can be generated? With a neat diagram showing the directions of current and forces derive an expression for the Hall voltage generated. | 6 | L2 | 4 | 1,2 |
| c) | Starting from the expression of conductivity for conductors obtain the expression for conductivity of n-type semiconductors. | 4 | L2 | 4 | 1,2 |
| d) | The Hall coefficient of a specimen of doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3 \text{ coulomb}^{-1}$; the resistivity of the specimen is $8.93 \times 10^{-3} \text{ ohm.m}$. Find the mobility and density of the charge carriers, assuming single carrier conduction. | 3 | L3 | 4 | 1,2 |
| 6. a) | How are bands formed? How does it explain the temperature dependence of conductivity in semiconductors? | 6 | L2 | 4 | 1,2 |
| b) | Why is a phonon needed for a photon emission in an indirect band gap semiconductor? Is this type of semiconductor suitable for making LEDs or lasers why? | 6 | L2 | 4 | 1,2 |
| c) | Explain the working of Solar cells. Define and explain the terms i) Efficiency ii) Fill factor. | 4 | L2 | 4 | 1,2 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|-----|
| 7. a) | A charged particle is moving in the positive x direction using an electric field, how can they be bent in any desired angle. Explain it quantitatively | 4 | L2 | 5 | 1,2 |
| b) | Describe the basic construction and working principles of a Scanning Electron Microscope (SEM). Provide a detailed explanation of the key components and their functions. | 6 | L2 | 5 | 1,2 |
| c) | Briefly explain Laue's method of X-ray diffraction. | 6 | L2 | 5 | 1,2 |
| 8. a) | What is the basic principle of Mass spectrometry? Explain two of its uses in biology. | 6 | L2 | 5 | 1,2 |
| b) | Derive Bragg's law for diffraction of X-Rays by crystals. | 6 | L2 | 5 | 1,2 |
| c) | Assume you have a crystal with a lattice spacing of 0.2 nm. If you shoot X-rays with a wavelength of 0.1 nm at the crystal and observe a diffraction angle (θ) of 30 degrees, what is the order (n) of the diffraction peak? | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
